ISO Standards and How the Chair Follows Them

Convertible Wheelchair-to-Bed Transfer System

The chair is a hand-operated wheelchair that is pushed by a caregiver. It also lies flat like a stretcher and moves the patient sideways onto a bed. The main standards for it are the **ISO 7176 series (Wheelchairs)**. Because it also lifts and moves a patient, a few general medical-device standards apply as well.

Meaning of the status column:

- **Met in design:** checked on the 3D model or by AI calculation.
- **Test on prototype:** can only be proven by testing the real chair.
- **Planned:** a choice or document that will be finalised with the prototype.

1. Wheelchair standards (ISO 7176 series)

Standard	What it says	How the chair follows it	Status
ISO 7176-1 Static stability	Measures how far the loaded chair can be tilted forward, backward and sideways before it tips over.	Tipping angles were worked out with a 120 kg patient, as a chair and lying flat, at the lowest and highest seat height. The worst case is about 20° (flat, at full height, tipping backward). As a chair at its lowest height, it takes 39° forward, 31° backward and 27° sideways. A wheelchair ramp is only about 5°. During a transfer the patient moves out past the wheels, so the chair is clamped to the bed frame first.	Met in design. Tilt test on prototype
ISO 7176-3 Brakes	Checks that the parking brakes keep the loaded chair from rolling on a slope.	All four 125 mm wheels have total-lock brakes: one pedal stops the wheel from rolling and from swivelling. Most wheelchairs brake only two wheels. The chair will be tested on a 7° ramp, the target commonly used for manual wheelchairs in Europe (EN 12183).	Test on prototype
ISO 7176-5 Size, mass and turning space	Sets how to measure and state the chair's outside size, its mass and the space it needs to turn around.	As a chair: 115 × 78 × 135 cm (length × width × height). Lying flat: about 181 cm long. Mass: about 79 kg. At 78 cm wide it passes through any door 80 cm or wider. The turning space will be measured on the prototype.	Met in design. Turning space on prototype
ISO 7176-7 Seat and wheel sizes	Sets how to measure seat height, depth and width, backrest angle and wheel sizes, for users up to 120 kg.	Seat 47 cm deep and 55 cm wide. Seat height 50 to 78 cm (hand crank). Backrest from upright to fully flat; the leg rest rises with it. Wheels: 125 mm casters. The chair is rated for 120 kg, the top of this standard's range.	Met in design

Standard	What it says	How the chair follows it	Status
ISO 7176-8 Strength: static, impact and fatigue	Pushes and pulls hard on the armrests, footrest, handles and frame, knocks the chair, and runs it on a rolling drum (200,000 cycles) and a kerb-drop rig (6,666 drops) to show that nothing breaks.	Every load-carrying part was checked by AI hand calculation, and the main parts also by FEA, for 1.5 times a 120 kg patient. The lowest safety factor is 2.3 (plastic gear teeth, where a failure only stops the belt). The frames and lift arms are at 2.7 to 3.0 or more. Everyday stresses are well below the limits for repeated loading. The impact, drum and kerb-drop tests need the real chair.	Static: met in design. Impact and fatigue: test on prototype
ISO 7176-11 Test dummies	Defines the dummies that stand in for a person during the tests above.	The prototype will be tested with dummy or sandbag loads up to the 120 kg maximum user mass, placed the way a patient's weight spreads over the back (49%), seat (39%) and legs (12%).	Test on prototype
ISO 7176-15 Manual, labels and information	Lists what the user manual and labels must tell the buyer: maximum user mass, sizes, warnings, use and care.	Label: maximum user 120 kg. The manual will cover: • the beds it fits: mattress top 47 to 75 cm high, mattress 9.5 to 28 cm thick, bed frame edge 2 to 10.5 cm deep • the safety steps in order: lock all four wheels, clamp to the bed, then transfer • cleaning and regular checks.	Planned
ISO 7176-16 Fire resistance of padding	Checks that padding and upholstery do not catch fire easily from a match or a cigarette.	The parts with padding or covers are the PU belts, the armrest pads and the leg pad cover. Fire-retardant grades will be bought, with the supplier's test certificate.	Planned
ISO 7176-22 Set-up before testing	Says how to set the chair to a fixed reference position before any test, so results can be repeated.	Reference set-up for testing: seat at its lowest (50 cm), backrest upright, both boards home and pinned, armrests fitted. Any other setting used in a test will be written in the test report.	Planned

2. Other standards for a patient-transfer device

Standard	What it says	How the chair follows it	Status
ISO 10535 Hoists for moving people	Safety rules for devices that lift and move a patient: strength, stability, safe lowering and caregiver effort.	The chair is not a hoist, but it lifts and moves a patient, so this standard is used as a guide: • The lift cannot drop: the screw is self-locking and holds wherever the crank stops. • The seat goes down only when the crank is turned. • The hardest crank turn feels like lifting about 12 kg. • The patient is never lifted off the surface; the belt carries them across.	Used as a guide

Standard	What it says	How the chair follows it	Status
ISO 14971 Risk management for medical devices	Find every hazard, judge how likely and how serious it is, reduce it, and check what risk is left.	Done as the design FMEA. Main examples: • tipping during transfer: bed clamp • backrest falling: gas spring lock, spring pin and stop blocks • boards sliding out: transit-lock pins • chair rolling: four total-lock wheels • seat sinking: self-locking screw.	Met in design. Update after testing
ISO 10993-1 Safety of materials touching the skin	Materials in contact with the skin must not cause irritation or allergy.	Skin-contact parts: PU belt, PU armrest pads, leg pad cover and powder-coated handles. Medical- or food-grade materials will be chosen, backed by supplier data. All surfaces are smooth and can be wiped clean with hospital disinfectant.	Planned
ISO 13485 Quality system for medical devices	How a maker must control design, purchasing and production of medical devices.	Applies when the chair goes into production. For the prototype, the parts list has revision control, every part has a drawing, and test results will be recorded.	Later (production)

3. Standards that do not apply

These parts of ISO 7176 are for electric wheelchairs. The chair has no motor, battery or electronics, so they do not apply:

- **ISO 7176-2** dynamic stability, **-4** battery range, **-6** speed, **-9** climate tests, **-10** obstacle climbing: all for electric wheelchairs.
- **ISO 7176-14** power and control systems, **-21** electromagnetic compatibility, **-25** batteries and chargers.

ISO 7176-19 (wheelchairs used as seats in vehicles) also does not apply. The chair is not meant to carry a patient in a car or ambulance, and the manual will say so.

4. Tests planned on the prototype

- Tilt test with 120 kg of load, as a chair and lying flat (ISO 7176-1).
- Parking brakes on a 7° ramp, fully loaded (ISO 7176-3).
- Measuring the size, mass and turning space (ISO 7176-5 and -7).
- Static loads on the armrests, footrest, handles and backrest (ISO 7176-8).
- Repeated cycles of lifting, reclining and transferring, to check for wear and loosening.

ISO standards are paid documents. The one-line summaries above are taken from the standards' public scope descriptions. We will try to get the full ISO standard documents while building the model.